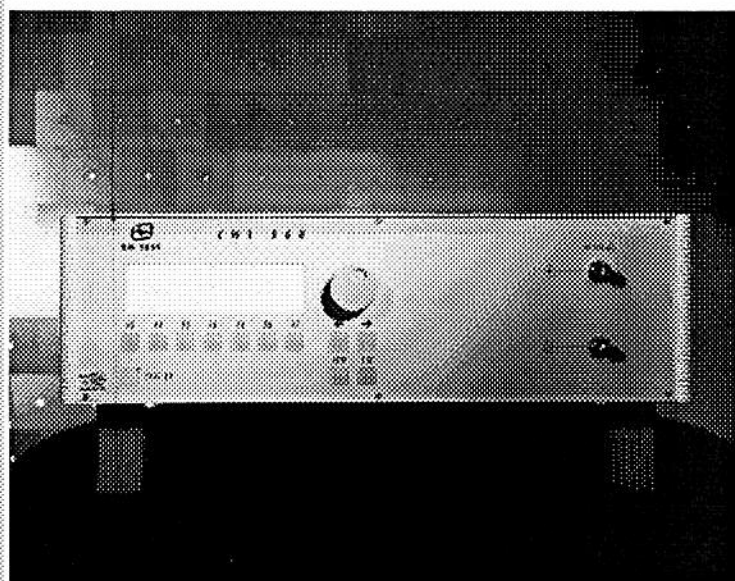


MANUAL

For Remote Control



Continuous Wave Simulator

Type CWS 500C

Remote Control for CWS 500C,
IEC 61000-4-6 / EN 61000-4-6

- IEC 61000-4-6
- EN 61000-4-6

emc test
equipment



Version: 1.00i01 / 05.9.02
Replaces: no previous version

Filename: manual remote CWS500C V420.doc
Printdate: 10.01.03



EM TEST
the difference

1. Remote Control

1.1. Interfaces

All following interfaces are standard features of the CWS 500C.

Serial RS 232 interface with 1200 - 19200 Baud

(8-databit, 1 start/stop bit)

25- Pin SubD PC / Printer	Signal		Signal	9- Pin SubD CWS 500C
2	TxD	----	RxD	3
3	RxD	----	TxD	2
7	GND	----	GND	5
	Screen		Screen	

Parallel IEEE 488 interface, addresses 1 - 30 selectable

- Command: (SH1, AH1, T4, L2, SR1, RL2, PP1, DC0, DT0, C0, E1)
- Connector and pin layout as per IEEE - 488 - 1975
- 24-pin Amphenol connector
- 8 ground pins

Equipment interface (15 pin Sub D)

The parallel equipment interface controls the external coupling networks.

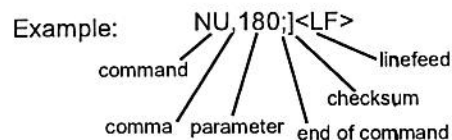
Printer

The printer may be connected to the serial RS 232 interface.

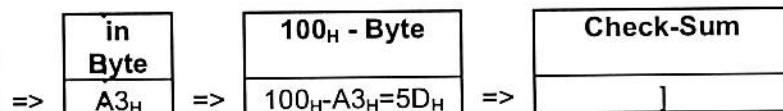
1.2. General information

The commands must be terminated by an <LF>. Just before the <LF> the check sum of the complete string must be transmitted.

Calculating : check sum $\neq 100_H$ - (sum of all ASCII codes in one byte)



	ASCII Hex
N	4E _H
U	55 _H
,	2C _H
1	31 _H
8	38 _H
0	30 _H
;	3B _H
SUM	1A3 _H



Remark:

sum of all ASCII codes in one byte:

Only the last 2 Digits of the sum of all ASCII codes in HEX will be considered.

The messages sent back by the CWS 500C are sent without check sum. At the end of the message there is also an <LF>.

Technical comments:

The CWS 500C-firmware is internally organized in 3 blocks.

Block 0: Setup (no relevance in remote mode)
Block 1: Test Routines
Block 2: Calibration

To access the desired program the correct block has to be set via remote commands (BS command).
To start the remote mode it is not necessary to switch to a default block.
After setting the CWS 500C to remote mode (CC;) it has to be checked which block is the actual one (BW command).

CWS 500C without Option PM250

Because of the lack of the current monitor several commands are not valid and the CN-command has to be treated as described below:

Command	Remarks
CN	The parameter BCI has to be set to 0. Otherwise the answer will be a string error (RR,10;)
KA KB KN KI AW	All these commands are not valid and will be answered with "RR,38;<LF>"

1.3. Parameters

Description	param.	Min - Max	Step	Unit	possible Values
Block	Block	0 = Setup 1 = Test Routines (incl. Up-/Download of the BCI stores) 2 = Calibration (incl. Up-/Download of the Calibration stores)			0 - 2
Voltage Displayed Voltage	V V _{disp}	0.1 - 1.0 1.2 - 4.0 4.5 - 7.0 8.0 - 20.0 22.0 - 60.0	0.1 0.2 0.5 1.0 2.0	V	1 - 600
Frequency	f	0.009 - 250.000	0.001	MHz	9 - 250000
Dwell time	td	0.3 - 9999,9	0.1	s	3 - 99999
Modulation	Mod	0 = OFF 1 = 2Hz 2 = 400Hz 3 = 1kHz 4 = Puls			0 - 4
Calibration store	Cal	0 = / 1 = F1 2 = F2 3 = F3 4 = F4			0 - 4
Pause	Pau	0 = The signal turns off after td 1 = The signal remains on after td			0 / 1
Transfer impedance store (BCI store)	BCI	0 = / 1 = BCI-Store 1 2 = BCI-Store 2			0 - 2
Transfer impedance	Imp	0.05 - 10.000	0.001	Ω	50 - 10000
Validity of the BCI-data	Valid_Name	0 = invalid data 1 = valid data			0 / 1
Name of the transfer impedance store	Name	max 8 ASCII-Characters			1 - 8 ASCII Characters
Date (16-Bit-integer)	Date	0 - 65535	1		0 - 65535
Time (16-Bit-integer)	Time	0 - 65535	1		0 - 65535
Channel of the ext. coupling matrix CNC508	Chn	0 - 8			0 - 8

1.4. C- Commands (Initialization)

Block	Comm.	Syntax	Description
all	CC	CC;	Check of the connection. - It is recommended to start every communication with the CWS 500C with this command. The answer can be used to check if the correct CWS 500C is connected to the PC. - Using the RS232 interface the CWS 500C will be set to remote mode. - The answer is as follows: CWS500C,CN,SWN,Type,Version;
			CN: 1 → CNC508 connected 0 → No CNC508 (Coupling Matrix)
			- SWN: SoftWareNumber - Example: 001015
			Type: W4 → without Option PM250 W2 → with Option PM250
			- Version: Firmware Version - Example: V4.32
1	CN	CN,V,Mod,f,Cal,td,Pau,BCI,Vdisp;	The CN-command handles the parameters for normal CWS 500C testing. - With Cal the internal stores F1-F4 can be selected. - If Pau is 0 the output will be switched off after the time td and (RR,00;<LF>) will be returned. - If Pau is 1 the output will stay switched on after the time td and (RR,00;<LF>) will be returned.
2	CL	CL,V,Mod,f;	The CL command handles the parameters for the calibration. - The calibration can be done only with the parameters U=10.0V, AM=OFF. - A limitation error will occur if the parameters are set to other values. For frequency all values are possible. - The CWS 500C sends back the voltage (RC,U;<LF>) to be set to get 10.0V calibration voltage at the output.

1.5. B- Command (Block manipulation)

Block	Comm.	Syntax	Description
All	BS	BS,Block;	Changes to the desired block. The answer is BS,Block;<LF>.
All	BW	BW;	Query of the actual block. The answer is BW,Block;<LF>.

1.6. A- Commands (start and stop a test)

Block	Comm.	Syntax	Description
1 + 2	AA	AA;	The AA command starts the test routine. At first all parameters and the routine must be handled by a CN- or a CL-command
1 + 2	AS	AS;	The AS command stops a running test.
1 + 2	AR	AR;	The AR command stops a running test and returns the CWS 500C to local mode. Note: This function is only available for the RS232 interface. The CWS 500C will answer with a string error (RR,10;<LF>) if it receives an AR-command through the IEEE-Interface.
1	AW	AW;	If a test stopped with the message RR,33;<LF> the test can be repeated with the AW-command but without current regulating.

1.7. N- Commands (changing single parameters)

Block	Comm.	Syntax	Description
1	NU	NU,V,V _{disp} ;	The NU command sends a new voltage level. V _{disp} is the value that will be displayed.
1	NF	NF,f;	The NF command sends a new frequency value.
1	NA	NA,Mod;	The NA command sends a new modulation type.
1	ND	ND,td;	The ND command sends a new value for the dwell time.
1	NK	NK,V,f, V _{disp} ;	The NK command sends a new voltage level and frequency value. V _{disp} is the value that will be displayed.

N-commands are only possible if no test is running. After receiving a N-command while a test is running the CWS 500C sends back a string error (RR,10;<LF>).

The output of the CWS 500C will only change after the AA-command.

Before sending a N-command to the CWS 500C for the first time the CWS 500C has to be initialized by a CN-command.

1.8. K- Commands (commands that have no direct affect to the test)

... commands that have no direct affect to the test)

Block	Comm.	Syntax	Description																																	
1	KA	KA,BCI;	Reads one of the BCI-stores (Upload to the PC) • "KA,number of elements;<LF>" will be returned, followed by all elements of the store. • Each element is followed by a <LF>. • For a description of the elements refer to the KB-command below and the KA-message in the table „Feedback Messages“.																																	
1	KB	KB,BCI; element0; element1; element2; ... element49; element 50; element 51;	Writes to one of the BCI-stores (transfer impedance of the current probe) (Download to the CWS 500C). • With BCI the store will be selected • See table parameters for the possible values of the impedances (Imp) • As values the current probes transfer impedances at the following frequencies have to be sent: <table> <tr> <td>9kHz – 10kHz</td> <td>Step 1kHz</td> </tr> <tr> <td>20kHz – 100kHz</td> <td>Step 10kHz</td> </tr> <tr> <td>200kHz – 1MHz</td> <td>Step 100kHz</td> </tr> <tr> <td>2MHz – 10MHz</td> <td>Step 1MHz</td> </tr> <tr> <td>20MHz – 100MHz</td> <td>Step 10MHz</td> </tr> <tr> <td>100MHz – 1GHz</td> <td>Step 100MHz</td> </tr> </table> • Not defined transfer impedances should be filled with the value of the nearest frequency where the transfer impedance is still defined. • All elements have to be sent with valid data every time. • The values are in Ω and not in $\text{dB}\Omega$. • The start and the stop frequency define the frequency range where the transfer impedance values are valid. • As answer after each element the CWS 500C will send back the element number starting at zero (each followed by a <LF>). • The old values in the BCI store will be overwritten only after correct receiving of all data. • The elements 49 - 51 are needed for control functions <table> <tr> <th>element</th> <th>description</th> <th>remarks</th> </tr> <tr> <td>0</td> <td>start frequency of the impedance</td> <td></td> </tr> <tr> <td>1</td> <td>stop frequency of the impedance</td> <td></td> </tr> <tr> <td>2 - 48</td> <td>impedance values</td> <td>Imp</td> </tr> <tr> <td>49</td> <td>invalid/valid</td> <td>65535d = valid</td> </tr> <tr> <td>50</td> <td>Min-value</td> <td></td> </tr> <tr> <td>51</td> <td>Max-value</td> <td></td> </tr> </table>	9kHz – 10kHz	Step 1kHz	20kHz – 100kHz	Step 10kHz	200kHz – 1MHz	Step 100kHz	2MHz – 10MHz	Step 1MHz	20MHz – 100MHz	Step 10MHz	100MHz – 1GHz	Step 100MHz	element	description	remarks	0	start frequency of the impedance		1	stop frequency of the impedance		2 - 48	impedance values	Imp	49	invalid/valid	65535d = valid	50	Min-value		51	Max-value	
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50	Min-value																																			
51	Max-value																																			
1	KN	KN,BCI,Date, Time,Name;	Sets a name for the BCI-store • The name consist of max. 8 ASCII-characters • If the name exists already "RR,32;<LF>" will be returned. • Sets a date and time for the BCI-store (both as 16bit integer) • The structure of date and time is not defined and can therefore be used as desired. • The storage of the date and time gives the possibility to control if the data in the BCI-store is actual.																																	
1	KI	KI,BCI;	Queries the name of the BCI-store in the CWS 500C																																	

2	KR	KR,Cal;	Reads one of the stores F1...F4 (Upload to the Computer). • "KR,number of elements;" will be sent back followed by all elements of the store. • See also the KR-message in the table „Feedback Messages“. • The values correspond to the voltage needed to fulfill the standard requirement at $V_0 = 10V$. That means programming the CWS 500C with these values results in a calibration voltage $V_{mr} = 1,66V$ which are required by the standard IEC61000-4-6. • During a test all voltages can be linearly calculated from these values. • The frequency range of the calibration data is from 100kHz up to 240MHz. • The frequency step size is 3% of the preceding frequency. • The elements 265 - 267 are needed for control purposes.		
			element	description	remarks
			0 - 264	calibration values (Cal-V)	voltage needed to reach the calibration level for 10V (1.66V) The range is the same as for V or V_{disp}
			265	invalid/valid	value = 255 => data are valid (all other values indicate that the data are invalid)
2	KW	KW,Cal; element0; element 1; element 2; ... element 265; element 266; element 267;	266	Min-value	Minimal calibration value
			267	Max-value	Maximal calibration value
2	KW	KW,Cal; element0; element 1; element 2; ... element 265; element 266; element 267;	Writes in a calibration store (download to the CWS 500C). • With Cal the store will be selected. • This command is used only to download previously uploaded calibration data. It is not intended to download self made data. • For a description of the elements refer to the KR-command above. • As answer after each element the CWS 500C will return the element number starting at zero (each followed by a <LF>). • All elements have to be sent with valid data every time. • The old values in the calibration store will be overwritten only after correct receiving of all data.		
All	KK	KK,Kop;	Sets a channel at the external coupling matrix CNC 508.		

1.9. Feedback Messages

Message	Description								
RR,00;<LF>	The test routine has been finished correctly.								
RR,03;<LF>	Calibration store is not valid.								
RR,04;<LF>	1. Clipping: corrected voltage is not within the limits. Output will be set to 60.0V 2. Calibration is not within the limits.								
RR,05;<LF>	Fail signal on Fail 1 The running test has been stopped by the CWS 500C								
RR,06;<LF>	Fail signal on Fail 2. The CWS 500C continues normally but it ignores all fail 2 signals until it receives a C-, a N- or an A-command.								
RR,10;<LF>	String error: there was an error in data transmission, which means too few or too many characters or the command is not acceptable.								
RR,11;<LF>	Test start is not possible or the test has been stopped because TEST ON is not switched on.								
RR,13;<LF>	No, or wrong coupling matrix connected. The required coupling mode can not be selected.								
RR,14;<LF>	One or several values are limited by the CWS 500C. The parameters are set to their minimum.								
RR,15;<LF>	Check sum error. The whole command should be sent again.								
RR,19;<LF>	Internal communication error. Turn off and on the CWS 500C.								
RR,20;<LF>	Not correctable limitation error. One or more parameter can not be set to its minimum.								
RR,25;<LF>	The received calibration data are corrupt (Min-value or Max-value are incorrect or Array-nr. 265 is not set to 255).								
RR,30, V_{corr} , max current, corr. current;<LF>	The test routine has been finished correctly using the current monitor. This answer will be returned if the test has been invoked by a CN-command with a valid BCI-store.								
	<table><tr><th>parameter</th><th>description</th></tr><tr><td>V_{corr}:</td><td>The factor between the programmed voltage and the reduced voltage (caused by an overcurrent).</td></tr><tr><td>max. current:</td><td>The highest measured current.</td></tr><tr><td>corr. current:</td><td>The reduced current if it was too high otherwise the highest measured current.</td></tr></table>	parameter	description	V_{corr} :	The factor between the programmed voltage and the reduced voltage (caused by an overcurrent).	max. current:	The highest measured current.	corr. current:	The reduced current if it was too high otherwise the highest measured current.
	parameter	description							
	V_{corr} :	The factor between the programmed voltage and the reduced voltage (caused by an overcurrent).							
max. current:	The highest measured current.								
corr. current:	The reduced current if it was too high otherwise the highest measured current.								
RR,31;<LF>	Answer to a CN-command if the selected BCI-store doesn't consist of valid data.								
RR,32;<LF>	Answer to the KN-command if - the desired name is already in use or - the data in the BCI-store is invalid. Answer to the KI-command with a non existing BCI-store								

RR,33, V _{corr} , max current, min current;<LF>	The current could not be reduced sufficiently => the test stopped.	
	The test can be repeated without control of the current by the AW-command.	
	parameter	description
	V _{corr} : max. current: min. current:	The factor between the programmed voltage and the reduced voltage (caused by an overcurrent). The highest measured current. The minimal measured current after reducing the output voltage to the minimum
RR,36;<LF>	Answer to a KA-command. There are no valid data in the selected BCI-store.	
RR,37;<LF>	The received values for the transfer impedance of the current probe (BCI) are not valid (Min-value or Max-value are incorrect or element-no. 49 is not set to 65535d)	
RR,38;<LF>	The sent command has no function because there is no PM250 (current monitor) built in.	
RR,Syn;<LF>	Confirmation after all 10sec (if td > 10s) that the test is still running normally.	
RC,U;<LF>	Internal calibration (answer to the CL-command). Returns the required output voltage to fulfill the standard requirement for Level 3/10V (the standard requires $V_m = 1.66V_{rms}$ at the Cal Input of the CWS 500C when calibrated with 10V). All test voltages other than 10V can be linearly calculated from the 10V-value.	
KI, Date, Time, Name, Valid_Data;<LF>	Returns the name of the BCI-store (Answer to the KI-command).	
	parameter	description
	Name:	Name which was stored by a KN-command
	Valid_Data:	Information if the data is valid in the BCI-store
KA, number of elements;<LF> Wert0;<LF> Wert1;<LF> Wert2;<LF> Wert51;<LF>	Date:	Date in 16-Bit-integer which was stored by a KN-command
	Time:	Time in 16-Bit-integer which was stored by a KN-command
	Answer to a KA-command.	
	- Returns the calibration-store. - Please refer to the KA-command for a detailed description of the data.	
KR, number of elements;<LF> Wert0;<LF> Wert1;<LF> Wert2;<LF> Wert267;<LF>	Answer to a KR-command.	
	- Returns the BCI-store. - Please refer to the KR-command for a detailed description of the data.	

In the case that no feedback message is necessary, after each received string (<LF>) the two letters of the command will be returned to check that the CWS 500C has understood the command.
If an error occurred one of the above messages will be returned without the two letters from the command.